Formula List for Diffraction

1. Grating formula

$$(a+b)\sin\theta = n\lambda$$

Where;

- $(a+b)$= Grating element
- θ = Angle of Diffraction
- n = order of Diffraction
- λ = Wavelength of wave getting Diffracted

2. Grating Element

$$(a+b)=\frac{1}{N}$$

Where;

- $(a+b)$= Grating element
- N = Number of lines per unit length of Grating

3. Resolving power of Grating (R.P.)

$$R.P.=\frac{\lambda}{d\lambda}=mN$$

Where;

- N = Number of lines per unit length of Grating
- m = order of Diffraction
- λ = Mean Wavelength of sources to be resolved
- d λ = Difference between Wavelength of sources to be resolved

DIFFRACTION PROBLEMS

Q1. A diffraction grating used at normal incidence gives a yellow line (λ =6000A^0)in a certain spectral order superimposed on a blue line (λ =4800A^0) of next higher order. If the angle of diffraction is sin^{-1}(3/4) calculate the grating element.

Given:- λ_1=6000A^0=6x10^{-5} cm; λ_2=4800A^0=4.8 x 10^{-5}cm ; $\theta=\sin^{-1}\left(\frac{3}{4}\right)$

Formula:- $(a+b)\sin\theta=n\lambda$; n=1,2,3,4 .....

<u>Solution:-</u> for given $(a+b)$ and θ ; n α 1/ λ

$$(a+b)\sin\theta=n\ \lambda_1$$

$$(a+b)\sin\theta=(n+1)\ \lambda_2$$

$$n\ \lambda_1=(n+1)\ \lambda_2$$

$$\frac{\lambda_1}{\lambda_2}=1+\frac{1}{n}$$

Therefore, $n=4$

$$(a+b)=\frac{n\lambda_1}{\sin\theta} = \frac{4\times 6x\ 10^{-5}}{\frac{3}{4}} = 32 \text{ x } 10^{-5} \text{ cm}$$

<u>Ans:- The grating element is 3.2 x 10^{-4} cm</u> .

Q2. Monocromatic light of wavelength 6560A falls normally on a grating 2 cm wide. The frist order spectrum is produced at an angle of 16°17’ from the normal. Calculate the total number of lines on the grating.

Given:- λ=6560A=6560 x 10^{-8}cm; width = 2cm ; n=1 ; θ=16.28°

Formula:- $(a+b)\sin\theta=n\lambda$

$$a+b = \frac{1}{(N)\, Number\ \ of\ \ lines\ \ per\ \ cm}$$

Total number of lines= N x width

<u>Solution:-</u> (a+b)= $\frac{n\lambda}{sin\theta} = \frac{6560}{\sin 16.28} \times 10^{-8}$ = 2.34×10^{-4} cm

Number of lines per cm = $\frac{1}{a+b}$ =4273

Total no. of lines = 4273 x 2 = 8547

<u>Ans:- There will be 8547 lines on the grating.</u>

Q3.A parallel beam of light is incident on a plane transmission grating having 3000 lines/cm. A third order diffraction is observed at 30®. calculate the wavelength of the line.

Given:-a+b = 1/3000 ; n=3 ; θ=30

Formula:- $(a+b)\sin\theta = n\lambda$, n=1,2,3…

<u>Solution:-</u> $\lambda = \frac{a+b}{n} \times sin\theta$

$$= \frac{1}{3000 \times 3} \times \sin 30$$

$$= \frac{1}{9000 \times 2} = 5.555\times10^{-5} \text{ cm}$$

<u>Ans:- The wavelength of the line is 5.555×10^{-5} cm.</u>

Q4. The visible spectrum ranges from $4000A^0$ to $7000A^0$. Find the angular breath of the first order visible spectrum produced by a plane grating having 6000 lines/cm when light is incident normally on the grating.

Given:- l1 = 4000A=4×10^{-5} cm l2= 7000A=$7\text{x}10^{-5}$ cm n=1 a+b=1/6000lines per cm

Formula:- $(a+b)\sin\theta = n\lambda$

Solution:- (a +b)sin $\theta_1 = \lambda_1$

$$\theta_1 = \sin^{-1}\frac{\lambda 1}{a+b} = \sin^{-1}(4x10^{-5}x\ 6000) = 13.88^{\circ}$$

(a +b)sin $\theta_2 = \lambda_2$

$$\theta_2 = \sin^{-1}\frac{\lambda 2}{a+b} = \sin^{-1}(7x10^{-5}x\ 6000) = 24.83^{\circ}$$

$$\theta_2 - \theta_1 = 24.83 - 13.88 = 10.95^{®}$$

Ans :- The Angular separation = 10.95®

Q5. In plane transmission grating the angle of diffraction for the second order principal maxima for the wavelength 5 x 10^{-5} cm is 35®. Calculate the number of lines/cm on the diffraction grating.

Given:- l=5x10^{-5}cm ; θ =35® ; n=2

Formula:- $(a+b)\sin\theta = n\lambda$; $\frac{1}{a+b}$ = number of lines/cm

Solution:- a+b=$\frac{n\lambda}{\sin\theta} = \frac{2\times5\times10^{-5}}{\sin 35°}$ = 1.74 x 10^{-4}

Number of lines per cm = $\frac{1}{a+b} = \frac{1}{1.74\times10^{-4}}$ = 5735

Ans:- The number of lines pre cm is 5735.

Q6. A grating has 620 ruling/mm and is 0.5 mm wide. What is the smallest wavelength interval that can be resolved in the third order at λ =481mm?

Given:- N = 620 x 0.5=310 ; λ =481x10^{-9}m ; m=3

Formula:- $\frac{\lambda}{d\lambda} = mN$

Solution:- d λ = $\frac{\lambda}{mN} = \frac{481\ \times10^{-9}}{3\times310}$ = 0.5172 x 10^{-9} m

$d\lambda = 0.5172\ A^{\circ}$

Ans:- The smallest wavelength interval is 0.5172 A°

Q7. Calculate the minimum number of lines required on grating that can just resolve the two sodium lines λ 1=5890A and λ 2=5893 A in third order.

Given:- λ_1=5890 x 10^{-8} cm, λ_2=5893 x 10^{-8}cm , m=3

Formula:- Resolving power $= \frac{\lambda}{d\lambda} = mN$

Solution:- $\lambda = \frac{\lambda_1 + \lambda_2}{2} = \frac{(5890+5896)10^{-8}}{2}$ = 5893 x 10^{-8} cm

d= (5896-5890)x10^{-8} = 6x10^{-8} cm

N = $\frac{\lambda}{m\ d\lambda} = \frac{5893 \times 10^{-8}}{3 \times 6 \times 10^{-8}}$ = 327

Ans:- Minimum of 327 lines are required on the grating.

Q8. Calculate the maximum order of diffraction maxima seen from plane transmission grating with 2500 lines per inch if light of wavelength 6900 A falls normally on it.

Given:- N=$\frac{1}{a+b} = 2500$lines/inch = 2500x 2.52 x 10^{-2} = 63 lines/m

λ= 6900A=6900x10^{-10} m

Formula:- $(a+b)\sin\theta = n\lambda$

Solution:- for n= n_{max}, sin θ =1

$n_{max} = \frac{a+b}{\lambda}$ = 2.3

Ans:- Maximum order of diffraction is 2

Single Slit and Double Slit

1. A slit of width 0.3mm is illuminated by a light of wavelength 5890Å. A lens whose focal length is 40 cm forms a Fraunhofer diffraction pattern Calculate the distance between first and the next bright image from the axis. (Ans: 0.0393 cm)

Diffraction Grating

1. In a plane transmission grating the angle of diffraction for second order principal maximum for the wavelength $5X10^{-5}$ cm is 30^0. Calculate the number of lines/cm of grating surface. (Ans: 5000)

2. In a plane transmission grating the angle of diffraction for the first order principal maximum is 20^0 for a wavelength 6500Å. Calculate the number of lines in one cm of the grating surface. (Ans: 5262)

Order of Spectrum

1. A plane transmission grating has 5000 lines/cm.
   a) Find out the highest order of spectrum observed if incident light has λ = 6000Å.

   b) If opaque spaces between the slits are exactly 2 times the transparent space and the maximum order observed is three, find which order of spectra will be absent.

   (Ans: 3^{rd} order; 3,6,9 order)

2. Calculate the maximum order of diffraction maxima seen from a plane diffraction grating having 5500 lines per cm if light of wavelength 5896Å falls normally on it. (Ans: 3)

3. The visible spectrum range from 4000Å to 5000. Find the angular breadth of the first observer visible spectrum produced by a plane grating having 6000 lines/cm when light is incident normally on the grating. (Ans: 3.57^0)

4. In a plane transmission grating, the angle of diffraction for second order principal maxima of wavelength $5X10^{-5}$ cm is 35^0. Calculate the number of lines/cm on diffraction grating.

Grating Element

1. A diffraction grating used at normal incidence gives yellow line (λ = 6000Å) in a certain spectral order superimposed on a blue line (λ = 4800Å) of the next higher order. If angle of diffraction is $\sin^{-1}(3/4)$, calculate the grating element. (Ans: $3.2X10^{-6}$)

Resolving Power

1. A plane grating just resolves two lines in the second order. Calculate the grating element if $d\lambda$ = 6Å, λ = $6X10^{-5}$ cm and width of ruled surface is 2 cm. (Ans: $4X10^{-3}$)

2. Find the minimum number of lines in a plane diffraction grating required to just resolve the sodium doublet (5890Å and 5896Å) in the (i)first order (ii)second order. (Ans: 982, 481)

3. A grating has 620 rulings/mm ans is 5.05 mm wide. What is the smallest wavelength interval that can be resolved in the third order at $\lambda = 481$ nm ? (Ans: $2.58X10^{-9}$)